LAB 10: Ordinary Differential Equation

Objective: To numerically obtain the solution of generic governing ODE of any physical system

Theory: equations, which are composed of an unknown function and its derivatives, are called differential equations. When the function involves one independent variable, the equation is called an *ordinary differential equation* (or ODE). Consider an ordinary differential equations of the form,

$$\frac{dy}{dt} = f(t, y)$$

Euler's Method –

New value = old value + slope × step size

in mathematical terms,

$$y_{i+1} = y_i + \phi h$$

where the slope ϕ is called an increment function. According to this equation, the slope estimate of ϕ is used to extrapolate from an old value yi to a new value yi+1 over a distance h. This formula can be applied step by step to trace out the trajectory of the solution into the future.

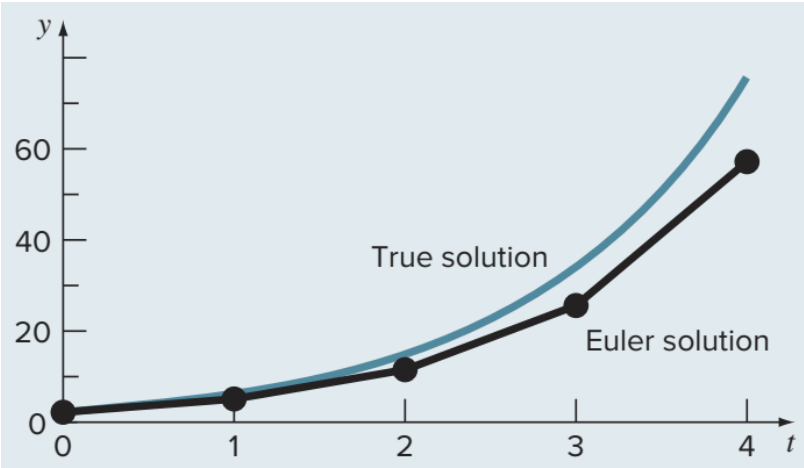
The first derivative provides a direct estimate of the slope at t_i

$\phi = f(t_i, y_i)$
where $f(t_i, y_i)$ is the differential equation evaluated at t_i and y_i.
$y_{i+1} = y_i + f(t_i, y_i)h$
This formula is referred to as *Euler's method* (or the Euler-Cauchy or point-slope method). A new value of y is predicted using the slope (equal to the first derivative at the original value of t) to extrapolate linearly over the step size h.

Problem Statement. Use Euler's method to integrate $y' = 4e^{0.8t} - 0.5y$ from $t = 0$ to 4 with a step size of 1. The initial condition at $t = 0$ is $y = 2$. Note that the exact solution can be determined analytically as

$$y = \frac{4}{1.3}(e^{0.8t} - e^{-0.5t}) + 2e^{-0.5t}$$

t	y_{true}	y_{Euler}	$\lvert\varepsilon_t\rvert$ (%)
0	2.00000	2.00000	
1	6.19463	5.00000	19.28
2	14.84392	11.40216	23.19
3	33.67717	25.51321	24.24
4	75.33896	56.84931	24.54



Assignment Problem –

Assignment –

Read about Modified Euler method and Attempt a numerical based on it.

One numerical as suggested by Faculty.

Sample numerical.

Solve the following initial value problem over the interval from $t = 0$ to where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dt} = yt^2 - 1.1y$$

(a) Analytically.
(b) Using Euler's method with $h = 0.5$ and 0.25.

Handwritten Notes

Exp - 10

* Euler's method = first order Runge Kutta

$$y_{i+1} = y_i + \phi h$$

h ← step size

ϕ ← slope at current point

* Modified Euler

Step by step:

1. predict next value using euler
2. compute slope at new prediction
3. Take avg of both slopes.

$$y_{n+1} = \frac{h}{2}\left[\phi_n + \phi_{n+1}\right] + y_n$$

$\phi_n \rightarrow \phi(x_n, y_n)$

$\phi_{n+1} \rightarrow \phi(x_n + h, y_n + h f(x_n, y_n))$

Ex: $\frac{dy}{dt} = yt^2 - 1.1y$

$$\therefore \frac{dy}{y} = (t^2 - 1.1)\,dt$$

$$\ln y = \frac{t^3}{3} - 1.1t + c$$

$$\therefore y = c_1 e^{t^3/3 - 1.1t}$$

$\because$ at $t = 0$ $y(0) = 1 \Rightarrow 1 = c_1$

$$\therefore y = e^{t^3/3 - 1.1t}$$

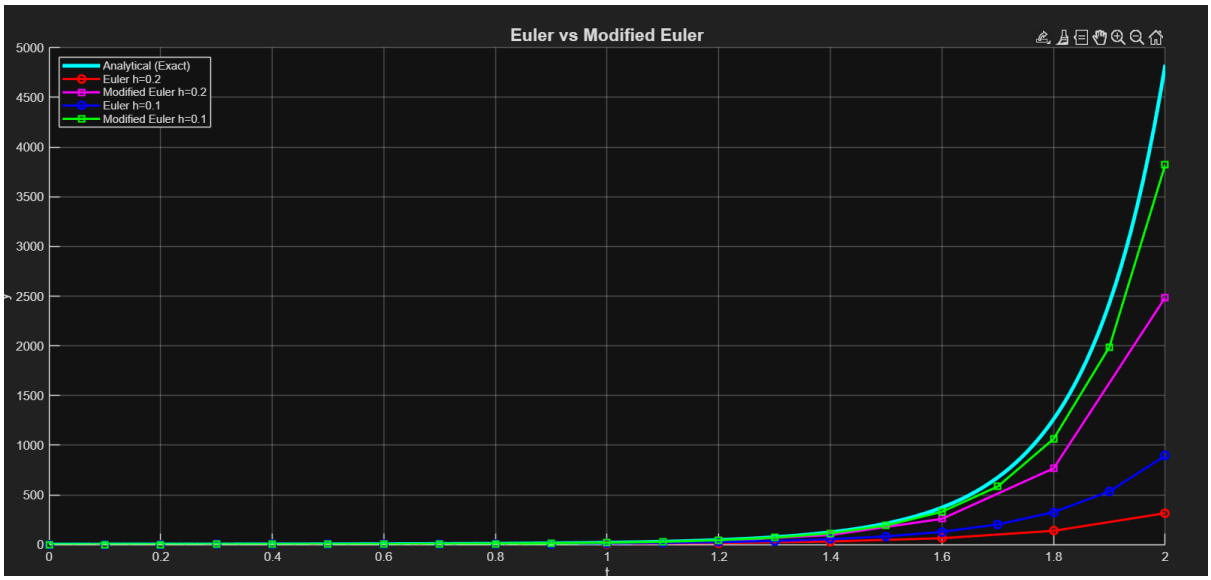
ex: $\frac{dy}{dt} = 2yt^{1.3} + 2.1y$

$\ln y = \frac{2}{2.3} t^{2.3} + 2.1t + c$

$\therefore y = c_1 e^{2/2.3\, t^{2.3} + 2.1t}$ //

$c_1 = 1$ //

Code and Output

```
clc
clear
close all

%analyticalsolution
t_exact = 0:0.01:2;
y_exact = exp((20/23)*t_exact.^2.3 + 2.1*t_exact);

% h=0.2
h = 0.2;
t1 = 0:h:2;

y_e1 = zeros(size(t1));
y_m1 = zeros(size(t1));

y_e1(1) = 1;
y_m1(1) = 1;

for i = 1:length(t1)-1

% Euler
    y_e1(i+1) = y_e1(i) + h*y_e1(i)*(2*t1(i)^1.3 + 2.1);

% Mod Euler
    k1 = y_m1(i)*(2*t1(i)^1.3 + 2.1);
    y_pred = y_m1(i) + h*k1;
    k2 = y_pred*(2*(t1(i)+h)^1.3 + 2.1);

    y_m1(i+1) = y_m1(i) + (h/2)*(k1 + k2);
end

%h = 0.1
h = 0.1;
t2 = 0:h:2;

y_e2 = zeros(size(t2));
y_m2 = zeros(size(t2));

y_e2(1) = 1;
y_m2(1) = 1;

for i = 1:length(t2)-1

% Euler
    y_e2(i+1) = y_e2(i) + h*y_e2(i)*(2*t2(i)^1.3 + 2.1);

% Mod Euler
    k1 = y_m2(i)*(2*t2(i)^1.3 + 2.1);
    y_pred = y_m2(i) + h*k1;
    k2 = y_pred*(2*(t2(i)+h)^1.3 + 2.1);

    y_m2(i+1) = y_m2(i) + (h/2)*(k1 + k2);
end


figure
hold on


plot(t_exact, y_exact, 'c', 'LineWidth', 3)


plot(t1, y_e1, 'r-o', 'LineWidth',1.8, 'MarkerSize', 6)
plot(t1, y_m1, 'm-s', 'LineWidth', 1.8, 'MarkerSize', 6)


plot(t2, y_e2, 'b-o', 'LineWidth', 1.8, 'MarkerSize', 6)
```

```
plot(t2, y_m2, 'g-s', 'LineWidth', 1.8, 'MarkerSize', 6)

grid on
xlabel('t','FontSize',12)
ylabel('y','FontSize',12)
title('Euler vs Modified Euler','FontSize',14)

legend('Analytical (Exact)', ...
       'Euler h=0.2','Modified Euler h=0.2', ...
       'Euler h=0.1','Modified Euler h=0.1', ...
       'Location','northwest')


```